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1. Greedy Approach 1

a.) Pseudocode:

Array is input

Sub array is array

```
While sub array is not empty {  
    Smallest_val = max int  
  
    For each element in array  
        If element < smallest_val  
            Smallest_val = element  
            Smallest_index = index  
    Return smallest_index  
    Output subarray[smallest]  
    sub array = array from smallest to end of sub_array  
}
```

b.) $O(n^2)$. Worst case iterates through n elements n times, then $n-1$ elements $n-1$ times, and so on, etc all the way down which when dropping lower order stuff, converts to n^2 .

c.) No, the following sequence fails: [4, 5, 6, 2, 3]. The algo takes 2, 3 as the longest subsequence, but the actual longest subsequence is 4, 5, 6.

2. Greedy Approach 2

a.) Pseudocode:

Array is input

longestArr = []

```
For (int i = 0; i < array.size(); i++)          n times  
    testArray = []  
    Testval = array[i]  
    testArray.append(testVal)  
    for(int j = i; j < array.size(); j++)        n-1, n-2, n-3 ... 0  
        j = find-min(j)                          n-1, n-2, n-3 ... 0  
        If (j is null)  
            j = array.size()  
        Else if(testVal > array[j])  
            testVal = array[j]  
            testArray.append(testVal)  
    If testArray.size > longestArr.size  
        longestArr = testArray
```

Return longestArr

Find-min(i):

For (int j = i+1; j < array.size(); j++)

 If array[j] > array[i]

 Return j

Return null

b.) Iterate through n-1 elements, n-1 times, then iterates through n-2 elements, n-2 times, and so on n times. This boils down to $O(n^3)$ runtime complexity when dropping lower order terms.

c.) No. The following set of numbers fails because it is greedy: 1 2 3 9 4 5 6 7 8. It sees the 9 after the 1-2-3 and takes it over the 4 because it is a smaller index. The iterations for subsequences look like such over the first 4 values:

1-2-3-4

2-3-9

3-9

9

Then, for position 4, we get 4-5-6-7-8 as our subsequence. This is the longest as no other subsequence after this will be that long(5-6-7-8 and the like). The real subsequence that should answer this problem is 1-2-3-4-5-6-7-8 but it never happens because 9 gets picked over 4.